

## Day 04: Inference for Prediction

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In groups: warm-up questions

Is education level associated with income? Researchers collected education level (in years) and income (in thousands of dollars) for a random sample of 32 employees working for the city of Riverview.

The researchers fit a simple linear regression of income on education and obtained the following output:

| term        | estimate | std.error | statistic | p.value |
|-------------|----------|-----------|-----------|---------|
| (Intercept) | 11.321   | 6.123     | 1.849     | 0.074   |
| education   | 2.651    | 0.370     | 7.173     | <0.001  |

**Q1.** Write the fitted regression equation.

**Q2.** Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

**Q3.** Interpret the intercept in the context of the problem. (Don't forget to specify units.)

A new researcher joined the team and decided that education should be standardized (to have mean 0 and SD 1) before fitting the regression model. The output from this regression is shown below:

| term             | estimate | std.error | statistic | p.value |
|------------------|----------|-----------|-----------|---------|
| (Intercept)      | 53.742   | 1.587     | 33.861    | <0.001  |
| scale(education) | 11.567   | 1.613     | 7.173     | <0.001  |

**Q4.** Write the fitted regression equation for this new model.

**Q5.** Interpret the slope coefficient in the context of the problem. (Don't forget to specify units.)

**Q6.** Interpret the intercept in the context of the problem. (Don't forget to specify units.)

**Q7.** Compare the two models. What's the same? What's different?

! Stop here

Let Amanda know when your group finishes!

### Two types of predictions

1. Predicting the **mean response** at a specific value of  $x$  (called a \_\_\_\_\_ interval)
  - e.g., the average leaf width in 2010
2. Predicting the response for a **specific future observation** (called a \_\_\_\_\_ interval)
  - e.g., the leaf width of *one specific leaf* in 2010

### Other Examples:

## Inference for prediction

Best point estimate:  $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$

General interval formula:

$$\text{estimate} \pm t^* \times SE$$

- $\hat{y}_0$  is our estimate
- use a  $t$ -distribution with  $df = n - 2$  to find  $t^*$

We'll need a *different* SE depending on whether we're building:

- a confidence interval for the mean response  $\hat{\mu}(Y \mid X_0)$ , or
- a prediction interval for an individual response  $Y \mid X_0$

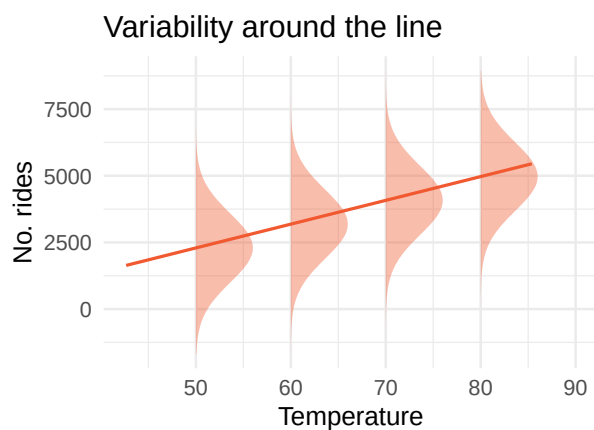
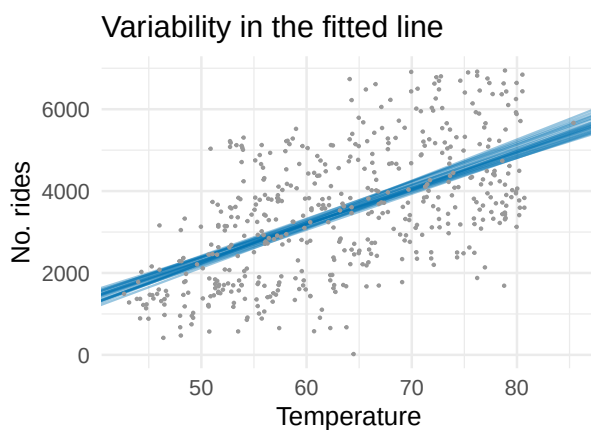
## Standard errors

$$SE(\hat{\mu}(Y \mid X)) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

$$SE(\hat{y}) = \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

- The interval for \_\_\_\_\_ will be wider
- As  $x_0$  gets farther from  $\bar{x}$ , both SE's get \_\_\_\_\_

$$SE(\hat{y}) = \sqrt{\underbrace{\hat{\sigma}^2}_{\text{variability in } \underline{\hspace{1cm}}} + \underbrace{\frac{\hat{\sigma}^2}{n} + \frac{\hat{\sigma}^2(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}_{\text{variability in } \underline{\hspace{1cm}}}}$$



## R commands

`predict()` quickly calculates  $\hat{y}$  for a given  $x$  (or a vector of  $x$ 's):

```
predict(nhanes_lm, newdata = data.frame(sedentary_hours = 10))
##          1
## 1.218353
predict(nhanes_lm, newdata = data.frame(sedentary_hours = c(8, 10, 12)))
##          1          2          3
## 1.155535 1.218353 1.281171
```

The `interval` argument specifies which type of interval you want:

```
predict(nhanes_lm, newdata = data.frame(sedentary_hours = 10),
  interval = "confidence")
##          fit          lwr          upr
## 1 1.218353 1.098451 1.338255
predict(nhanes_lm, newdata = data.frame(sedentary_hours = 10),
  interval = "prediction")
##          fit          lwr          upr
## 1 1.218353 -0.520678 2.957384
```

The `level` argument sets the confidence level (default 0.95):

```
predict(nhanes_lm, newdata = data.frame(sedentary_hours = 10),
  interval = "prediction", level = .99)
##          fit          lwr          upr
## 1 1.218353 -1.069557 3.506263
```

Adding `se.fit = TRUE` returns the standard error needed for “by hand” calculations:

```
predict(nhanes_lm, newdata = data.frame(sedentary_hours = 10),
  se.fit = TRUE)
## $fit
##          1
## 1.218353
##
## $se.fit
## [1] 0.06106312
##
## $df
## [1] 657
##
## $residual.scale
## [1] 0.8835349
```

## In groups: prediction and confidence intervals

### Tip

The R code for the following questions is found at  
<https://aluby.github.io/stat230/activities/04R-prediction>  
The URL is also posted on Moodle.

Now, we'll explore the `city` data using R.

**Q1.** Make a plot of `income` against `education` and overlay the linear regression equation. Does it look like what you expected?

**Q2.** Make a plot of `income` against `scale(education)` and overlay the linear regression equation. Does it look like what you expected?

**Q3.** Find the linear regression equation from Q1. Confirm that it matches the table at the beginning of this worksheet.

**Q4.** The first person in this dataset has `education = 8`. Calculate the expected income of this person using the first regression equation.

**Q5.** If we want to predict the income of **this** person, should we use a confidence interval or a prediction interval?

**Q6.** Use R to construct the appropriate 89% interval for this person's income. Record this interval below.

**Q7.** Interpret the interval in context.

**Q8.** Run the code to produce a scatterplot, regression line, and both types of intervals. Which is the prediction interval and which is the confidence interval? How can you tell?